Sensor applications

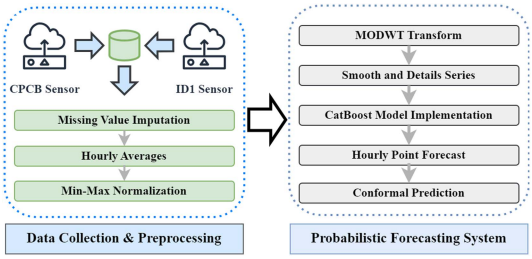
A Novel Hybrid Approach For Efficiently Forecasting Air Quality Data

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Abstract—Accurate and reliable air quality forecasting is essential for protecting public health, sustainable development, pollution control, and enhanced urban planning. This letter proposes a novel architecture namely wavelet-based CatBoost to forecast the real-time concentrations of air pollutants by combining the maximal overlapping discrete wavelet transform with the CatBoost model. This hybrid approach efficiently transforms time series of air pollution concentration levels into high-frequency and low-frequency components, thereby extracting signal from noise and improving prediction accuracy and robustness. Evaluation of two distinct regional datasets, from the Central Air Pollution Control Board sensor network and a low-cost air quality sensor system, underscores the superior performance of our proposed methodology in real-time forecasting compared to the state-of-the-art machine learning and deep learning architectures.



Index Terms—Sensor applications, air quality, CatBoost, conformal prediction, wavelet analysis, real-time forecasting.

I. INTRODUCTION

Air pollution threatens public health and the environment worldwide. Rapid industry and urbanization have increased air pollution, making sustainable living difficult. The WHO provided guidelines for six major air contaminants and created National Ambient Air Quality Standards to address these problems. These pollutants, including nitrogen dioxide (NO_2), ozone (O_3), carbon monoxide (CO), sulfur dioxide (SO_2), particulate matter with diameters of 2.5 mm or less ($PM_{2.5}$), and 10 mm or less (PM_{10}), originate from various sources, and natural processes. Despite considerable efforts to reduce emissions and lower ambient concentrations of these pollutants, air pollution remains a significant cause of mortality worldwide.

To mitigate these concerns related to polluted air, several countries have implemented real-time air quality forecasting systems [1]. Accurate prediction of air pollutant concentration levels is paramount for public awareness campaigns, environmental management, and healthcare interventions, among many others. In recent years, machine learning algorithms have emerged as powerful tools for enhancing the accuracy of air quality predictions [2]. These computational techniques, driven by their ability to decipher complex patterns and relationships within vast datasets, offer a promising avenue for advancing pollutant forecasting models [3], [4], [5].

However, existing methods encounter challenges adapting to diverse monitoring environments and accurately predicting pollutant concentrations in real-time [6]. Traditional models often struggle with the dynamic nature of air quality, hindered by limitations in handling nonlinear relationships and nonstationary variations in pollutant dynamics. To address these challenges, wavelet-based transformation is widely applied in other domains, including epidemiology [7] and economics [8], [9]. These wavelet-based forecasting approaches transform nonstationary seasonal data into a series of uncorrelated distinct components. Subsequently, the component series are modeled with various statistical and deep learning [10] architectures to solve diverse problems in time series forecasting. While the accuracy of these wavelet-based deep learning architectures surpass traditional approaches for ultra-long time series, their training and prediction times obstruct the generation of real-time forecasts. To mitigate this challenge, decision tree-based boosting architectures are favored for their faster convergence time in the presence of large datasets [11]. Several boosting algorithms, including random forest, extreme gradient boosting, and light gradient boosting machines, offer higher accuracy compared to deep learning methods, but they overlook the sequential nature of the time series, resulting in target leakage [12]. The recently developed CatBoost algorithm employs an ordered boosting methodology to overcome this issue [13].

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Data and codes are available online at https://github.com/jborah2/WaveCatBoost-Time-Series-Model.
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In this letter, we develop a novel wavelet-based CatBoost (WaveCatBoost) model for generating accurate real-time forecasts of air pollutant concentration levels. This integration aims at leveraging a robust and versatile model capable of delivering reliable point and probabilistic forecasts across a range of air pollutants via deploying a conformal prediction approach. By conducting an extensive experimental evaluation and comparing existing models, we demonstrate the superiority of our proposed approach in advancing air quality forecasting and addressing the critical gaps present in the current literature. The rest of this letter is organized as follows. Section II first illustrates the data collection and preprocessing phases, and then it details the proposed approach. Section III shows the achieved results and the comparative analysis. Finally, Section IV concludes this letter.

II. MATERIAL AND METHODOLOGY

A. Data and Preprocessing

Two datasets from sensors located in Meghalaya, India, region are used in this study. One of the datasets is from the Central Pollution Control Board (CPCB), and the other is collected using a low-cost air quality sensor (ID1). The raw data are spaced in 15 min intervals and consisted of missing values. Data preprocessing, which includes handling missing values and normalizing data, is crucial for improving the accuracy and efficiency of machine learning models. To address this, we employed imputation that ensures that the continuity of the time series data are preserved without introducing significant bias. We addressed missing data and calculated hourly averages, followed by min–max normalization to optimize the convergence of forecasting algorithms. Normalization is a key preprocessing step, especially when dealing with machine learning models that are sensitive to the scale of input features. We applied min–max normalization to rescale the pollutant concentration levels to a range of [0, 1]. This not only improves the convergence speed of the model during training but also ensures that each feature contributes equally to the prediction process. Without normalization, features with larger numerical ranges could dominate the model training, leading to suboptimal performance. The data used extends from July 2022 to July 2023. Four train-test scenarios with forecast horizons of 1, 7, 14, and 31 days are considered here.

B. Proposed WaveCatboost

The key concept behind the WaveCatBoost framework involves employing the scale-invariant maximal overlapping discrete wavelet transformation (MODWT) [14] to denoise the air pollutant concentrations series and modeling them using an ensemble of CatBoost models. The MODWT decomposition is applied to filter the normalized air pollutant data $(Y_t)_{t=1}^{N}$ using a sequence of wavelet $\{p_m; m = 0, 1, \ldots, \mathcal{M}-1\}$ and scaling $\{q_m; m = 0, 1, \ldots, \mathcal{M}-1\}$ filters. These filters adhere to the even-length scaling assumption $\sum_{m=0}^{\mathcal{M}-1} p_m p_{m+2n} = \sum_{m=0}^{\mathcal{M}-1} q_m q_{m+2n} = 0$ for any nonzero integers n. Consequently, applying the pyramid algorithm [14] generates wavelet and scaling coefficients for the transformed time series, allowing the original series Y_t to be represented as a set of equal-sized uncorrelated time series

$$Y_t = \sum_{k=1}^{K} D_{k,t} + S_{K,t} \quad t = 1, 2, \ldots, N \tag{1}$$

where $K + 1 = \lfloor \log_e N \rfloor$ is the number of decomposition levels applied, $D_{k,t} (k = 1, 2, \ldots, K)$ is the kth details series of Y_t, capturing the local fluctuations of the series, and the smooth series $S_{K,t}$ preserves the low-frequency trend of Y_t. This iterative decomposition essentially helps filter the true signal from the noisy data and aids in modeling long-term dependent, nonstationary, and seasonal data by breaking it into several lower-resolution components. Subsequently, the task of generating h-step ahead forecasts of Y_t based on its historical data can be solved by forecasting these low-resolution components based on their corresponding prior observation. In our proposed WaveCatBoost architecture, we generate the forecasts for each of the wavelet and scaling coefficients of Y_t using the gradient boosting CatBoost [13] algorithm and ensemble them using an inverse MODWT (IMODWT) [14] approach as

$$\hat{Y}_{N+h} = \text{IMODWT}\left(\hat{D}_{1,N+h}, \ldots, \hat{D}_{K,N+h}, \hat{S}_{K,N+h}\right) \tag{2}$$

with $\hat{D}_{k,N+h} = f(D_{k,1}, D_{k,2}, \ldots, D_{k,N})$; $k = 1, 2, \ldots, K$ and $\hat{S}_{K,N+h} = f(S_{K,1}, S_{K,2}, \ldots, S_{K,N})$, where the function f indicates the CatBoost model applied on each of the p lagged values of details and smooth series. The CatBoost framework operates by iteratively combining weak symmetric decision trees into a strong regressor. Unlike traditional gradient boosting methods, it is prone to prediction shifts from target leakage. During the learning process, the training data undergoes a series of time-dependent permutations, denoted as $(\sigma_0, \sigma_1, \ldots, \sigma_u)$. The initial permutation σ_0 determines the leaf values, while the subsequent u permutations decide the structure of the weak learners. At each iteration v, the algorithm considers a randomly chosen permutation σ_r. The supporting models $M_{r,j}$ are then fitted to the preceding j observations of σ_r, generating the output $M_{r,j}(i)$ for the ith instance in σ_r. To construct the tree T_v at the current iteration, gradients for the ith

Algorithm 1: WaveCatBoost model

Require: Air pollutant concentration data $\{Y_t\}_{t=1}^{N}$
Ensure: h-step ahead forecast $\{\hat{Y}_{N+1}, \ldots, \hat{Y}_{N+h}\}$
Initialize the MODWT algorithm with Haar filter and $K = \log_e N$ levels.
Transform the training data $\{Y_t\}$ using the MODWT approach into K uncorrelated details series $\{D_{k,t}\}_{k=1}^{K}$ and a smooth series $\{S_{K,t}\}$
for $k = 1, 2, \ldots, K$ **do**
  Apply CatBoost model to the kth detail series $D_{k,t}$.
  $\sigma_0, \sigma_1, \ldots, \sigma_u \leftarrow$ Time-dependent permutations of $D_{k,t}$.
  σ_0 decides the leaf nodes and the next u permutations determine the structure of the symmetric weak learners.
  for $v = 1, 2, \ldots, V$ **do**
    $\sigma_r \longleftarrow$ random permutations of $[\sigma_0, \sigma_1, \ldots, \sigma_u]$.
    Fit $M_{r,j}$ to j preceeding observations of σ_r and generate output $M_{r,j}(i)$ for i^{th} instance of σ_r.
    Calculate leaf values of the week learner T_v by averaging the gradients $\text{grad}_{r,\sigma_r(i)-1}$ (3) of previous examples belonging to the same node $\text{Leaf}_{r(i)}$.
    Boost the models $M_{r',j}$ of $\sigma_{r'}$; $r' \neq r$ using T_v.
  end for
  Calculate leaf value for the CatBoost model by applying gradient boosting approach to V week learners' leaf nodes and generate the one-step ahead forecast for $D_{k,t}$.
  Iteratively generate multi-step ahead forecast.
end for
Similarly, utilize a CatBoost architecture on $S_{K,t}$ to generate the corresponding one-step ahead and multi-step ahead forecast iteratively.
Apply Inverse MODWT transform on the component forecasts from the ensemble of CatBoost models and generate the final forecasts of the desired horizon as in (2).

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Table 1. Performance of the Proposed WaveCatBoost Model and Selected Baseline Forecasters for CPCB and ID1 Sensors, Evaluated Based on the MASE Metric for Different Forecast Horizons

Pollutant	Horizon	CPCB Sensor								ID1 Sensor							
		RNN	LGBM	TCN	TFT	Transformer	XGB	CatBoost	WaveCatBoost	RNN	LGBM	TCN	TFT	Transformer	XGB	CatBoost	WaveCatBoost
NO_2	1d	2.45	0.79	1.45	1.41	3.28	0.91	0.82	**0.57**	2.05	1.27	2.15	1.62	2.07	1.40	1.20	**0.91**
	7d	1.35	**0.66**	0.67	2.65	0.93	**0.66**	1.04	0.84	1.66	0.82	1.64	1.76	1.76	0.95	0.84	**0.52**
	14d	0.81	**0.71**	1.79	4.27	2.05	0.72	0.73	0.72	1.57	0.85	1.56	2.06	1.63	0.92	1.69	**0.42**
	31d	1.98	0.75	1.87	1.18	1.35	0.76	1.17	**0.47**	1.47	1.15	1.48	1.95	1.86	1.14	1.36	**0.40**
O_3	1d	5.19	1.87	1.85	2.43	11.47	1.52	2.69	**0.98**	3.65	1.61	3.32	2.79	3.75	1.72	1.49	**1.42**
	7d	5.46	1.68	4.66	6.16	5.82	1.51	4.84	**0.76**	3.57	1.64	2.55	2.84	3.18	1.96	1.79	**0.72**
	14d	6.24	1.23	4.38	14.68	9.54	1.27	1.77	**1.11**	3.85	1.80	3.67	3.74	3.12	2.06	2.32	**0.64**
	31d	3.56	2.46	4.26	2.71	3.56	2.44	5.37	**0.25**	3.59	2.72	3.63	3.56	3.61	2.87	2.66	**1.11**
CO	1d	2.36	3.36	2.90	2.45	**0.97**	3.21	2.87	2.69	4.47	2.48	3.82	2.95	4.47	2.54	2.32	**1.53**
	7d	3.74	3.62	3.94	5.29	3.74	3.59	3.57	**2.61**	2.85	1.52	2.64	2.95	2.84	1.46	1.36	**0.79**
	14d	2.76	3.06	2.79	4.34	2.77	2.88	3.15	**1.63**	3.08	1.93	3.08	3.71	3.06	1.83	3.36	**0.83**
	31d	1.92	2.05	1.96	3.29	1.91	1.98	2.03	**0.61**	2.79	2.58	2.79	3.48	3.57	2.22	3.26	**0.71**
SO_2	1d	34.35	11.46	28.11	**7.76**	40.15	10.62	9.35	23.93	1.81	0.45	0.85	0.49	1.79	**0.43**	0.44	1.31
	7d	16.83	15.43	14.62	20.72	11.59	14.39	**5.84**	7.30	2.36	0.34	2.23	0.61	2.21	0.33	**0.30**	0.61
	14d	10.70	7.35	9.37	24.75	9.57	8.14	11.31	**6.17**	2.15	0.22	2.82	1.66	3.18	0.26	**0.24**	0.58
	31d	9.06	11.22	10.34	11.44	10.79	9.13	9.81	**5.33**	2.35	0.41	2.95	0.81	1.88	**0.26**	0.31	1.31
$PM_{2.5}$	1d	2.88	3.81	**2.50**	3.52	6.57	2.95	3.02	5.06	7.31	4.67	7.03	4.68	7.35	5.53	4.45	**0.85**
	7d	6.94	6.86	7.05	7.38	6.69	5.92	6.57	**5.07**	7.09	3.01	5.22	4.22	5.56	2.65	2.43	**0.58**
	14d	4.40	6.98	4.40	7.07	4.41	6.20	6.30	**2.34**	8.20	4.43	8.71	7.10	7.39	3.67	5.58	**0.69**
	31d	2.73	2.85	2.83	4.41	2.73	2.83	2.83	**1.57**	7.21	5.26	7.01	5.97	5.05	4.49	5.29	**0.42**
PM_{10}	1d	**1.01**	6.69	1.79	6.37	4.16	3.81	5.77	3.12	7.31	4.34	7.06	4.24	7.35	5.59	4.22	**0.86**
	7d	5.20	5.30	5.32	6.19	5.11	4.99	5.24	**3.70**	7.37	2.83	5.42	4.34	5.81	2.46	2.27	**0.57**
	14d	3.53	4.59	3.55	5.90	3.54	4.36	4.89	**1.48**	8.48	4.29	9.08	7.08	7.65	3.29	5.44	**0.67**
	31d	2.39	2.40	2.51	3.64	2.37	2.43	2.40	**0.98**	7.57	5.39	7.35	6.09	5.36	4.50	5.47	**0.41**

time-index are computed by considering previous time steps in the σ_r permutation, i.e., $\text{grad}_{r,\sigma(i)-1}(i)$, where

$$\text{grad}_{r,j}(i) = \left.\frac{\partial L\left(y_i, A\right)}{\partial A}\right|_{A=M_{r,j}(i)}. \tag{3}$$

The leaf value for the ith instance, used for evaluating candidate splits, is obtained by averaging the corresponding gradients $\text{grad}_{r,\sigma_r(i)-1}$ of preceding examples belonging to the same leaf node $\text{Leaf}_{r(i)}$. Once the tree T_v is constructed, it is used for boosting all the supporting models $M_{r',j}$ with varied sets of leaf values depending on the first j instances in the $\sigma_{r'}$ permutation. When all the weak learners are constructed, the leaf value of the CatBoost model is calculated by the standard gradient boosting procedure. For generating the forecasts $\hat{D}_{k,N+h}$; $k = 1, 2, \ldots, K$ and $\hat{S}_{K,N+h}$, the leaf values are computed based on target statistics learned from the CatBoost architecture using lagged inputs. The algorithmic structure of the proposed WaveCatBoost model is given in Algorithm 1.

C. Probabilistic Forecasting Using Conformal Prediction

Along with the point forecasts generated by the WaveCatBoost model, we focus on quantifying the uncertainty associated with these forecasts using the conformal prediction approach [15]. This non-parametric procedure generates a probabilistic band around the point forecasts based on a conformal score (CS_t). To calculate the CS_t corresponding to Y_t [16], the architecture considers p lagged values of the target series (Y_{t-p}) and applies the WaveCatBoost and an uncertainty model $\hat{\mathbf{U}}$ such that $CS_t = \frac{|Y_t - \text{WaveCatBoost}(Y_{t-p})|}{\hat{\mathbf{U}}(Y_{t-p})}$. Following the sequential nature of Y_t and the CS_t, the conformal quantile can be computed with a weighted aggregation method having a fixed window $w_t = \mathbb{1}(t' \geq t - \kappa), \forall\, t' < t$ of length κ as

$$CQ_t = \inf\left\{q : \frac{1}{\min\left(\kappa, \text{t'-1}\right)} \sum_{t'=1}^{t-1} CS_{t'} w_t \geq 1 - \alpha\right\}. \tag{4}$$

The conformal prediction intervals based on these weighted conformal quantiles can be estimated as

$$\left[\text{WaveCatBoost}\left(Y_{t-p}\right) \pm CQ_t \hat{\mathbf{U}}\left(Y_{t-p}\right)\right]. \tag{5}$$

III. EXPERIMENTAL RESULTS

This section presents the experimental results showcasing the efficacy of our proposed WaveCatBoost framework compared to benchmark forecasting methods. In our analysis, we compare the performance of the proposed WaveCatBoost architecture with state-of-the-art forecasting methodologies including light gradient-boosting machines (LGBM), extreme gradient-boosting (XGB), Transformer, temporal convolution network (TCN), recurrent neural network (RNN), temporal fusion transformer (TFT), and CatBoost model [6]. We measure the forecast accuracy of various frameworks using the mean absolute scaled error (MASE) metric, one of the most recommended performance measures in forecasting literature, on the test data as follows:

$$\text{MASE} = \frac{N - S}{h} \frac{\sum_{t=N+1}^{N+h} \left|\hat{Y}_t - Y_t\right|}{\sum_{t=S+1}^{N} \left|Y_t - Y_{t-S}\right|} \tag{6}$$

where h is the forecast horizon, N is the length of training data, $\hat{Y}_t$ is the forecast of Y_t at time t, and S is the seasonality of the data.

Table I displays the performance of selected forecasting architectures, incorporating only those that perform in at least one scenario. For the CPCB sensor, the MASE metric values in the table indicate that the WaveCatBoost model outperforms all the baseline architectures for long-range forecasting periods of 14 and 31 days for most air pollutants. This improved performance is attributed to the hybrid training approach, where MODWT aids in extracting signals from noise, and CatBoost accurately extrapolates these transformed signals. In the 7-days forecasting scenario, WaveCatBoost performs comparably to other boosting methods, such as LGBM, XGB, and CatBoost, across various pollutants. Furthermore, for short-term forecasts with a 1-day horizon, WaveCatBoost demonstrates competitive performance with transformer-based architectures. In addition, the MASE metric for ID1 sensors in Table 1 indicates that our proposed model outperforms all baseline models across different horizons for all air pollutants except for SO2, where XGB and CatBoost models offer competitive results.

Furthermore, we study the statistical significance of the improvement in model performance using multiple comparisons with the best (MCB) test procedure [17]. This distribution-free test ranks each model based on its MASE metric across different air pollutants and computes their average rank. The model with the lowest MASE metric is identified as the "best" method, determined by achieving

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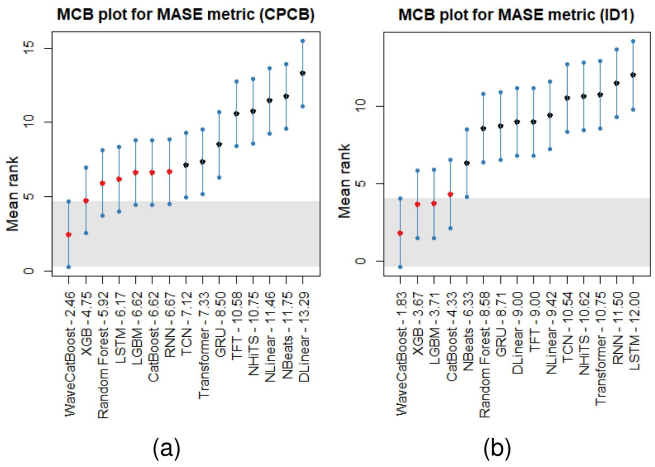


Fig. 1. MCB plots for MASE metric of the various forecasting models across all forecast horizons utilizing two different sensors (a) CPCB and (b) ID1.

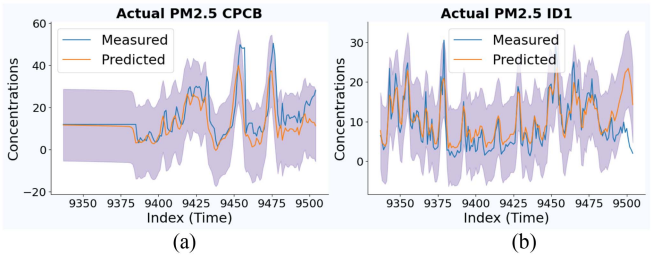


Fig. 2. Actual versus Predicted plot for $PM_{2.5}$ with upper and lower boundary show 7 day conformal prediction interval plot on (a) CPCB and (b) ID1 datasets.

the minimum rank. The MCB plots depicted in Fig. 1 illustrate that WaveCatBoost models exhibit superior performance for both the ID1 and CPCB sensors. In addition, the boundary of the critical distance (blue line) of the best-performing model serves as the reference value (shaded region) for the test. Given that most of the baseline forecasters have nonoverlapping critical distance (blue line) with the reference value of the test, we can conclude that their performance is significantly inferior to that of the proposed WaveCatBoost architecture. The overall experimental analysis underscores the effectiveness of our proposed model in outperforming a diverse range of benchmark models, emphasizing its potential as a reliable tool for real-time air quality forecasting. To quantify the uncertainty in the WaveCatBoost forecasts, we utilize the conformal prediction approach as in (5). The point forecasts generated by the WaveCatBoost model, ground truth, and the conformal prediction intervals at $\alpha = 0.05$ of different forecast horizons for each sensor data, respectively. Probabilistic forecasts of 7 days-ahead forecasting horizon for both the datasets are depicted in Fig. 2. It is evident that the proposed framework is generalizable and robust, offering valuable insights for environmental monitoring and public health interventions.

IV. CONCLUSION

In this letter, we designed a novel method, namely, WaveCatBoost model to forecast real-time air pollutant concentration levels. Our architecture integrates wavelet decomposition with the CatBoost approach to effectively capture the nonstationary and long-term dependencies inherent in pollutant time series data. Experimental evaluation using two real-world datasets from Meghalaya, India, demonstrates that our method outperforms baseline forecasting methods. Furthermore, we assess the robustness of the model through statistical significance tests and provide probabilistic bands for our forecasts based on a conformal prediction approach. These findings are crucial for advancing the field of air quality forecasts and guiding future research endeavors to enhance predictive models for environmental sustainability. These findings highlight the potential of the WaveCatBoost model to overcome the limitations of existing forecasting methodologies. In light of our results, As a future direction, further analysis of the WaveCatBoost architecture in forecasting air pollutant levels will consider spatial depencies in air quality research.

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